

Lesson 16: Reusable Modules

Learning Objectives

- Design modules for reusability across projects
- Use module outputs to expose complex data
- Create a module registry-compatible module
- Use the Terraform Registry for public modules

1. Characteristics of a Reusable Module

A well-designed reusable module follows these principles:

Principle	Description
Minimal required inputs	Only require what is truly unique to each call
Sensible defaults	Optional parameters have good defaults
Complete outputs	Expose all useful attributes callers may need
No hardcoded values	Everything configurable via variables
Consistent naming	Use <code>this</code> as the resource name, not project-specific names
Documentation	<code>README.md</code> explaining usage

Naming Convention

Modules should avoid hardcoded prefixes in resource names:

```
# Bad – hardcoded project name in module
resource "aws_s3_bucket" "this" {
  bucket = "myproject-${var.bucket_name}" # can't be reused by other
  projects
}

# Good – let the caller control the name
resource "aws_s3_bucket" "this" {
  bucket = var.bucket_name # caller provides full name
}
```

2. Module Pattern: S3 Bucket

```
# modules/s3-bucket/variables.tf
variable "bucket_name" {
```

```
    description = "Name of the S3 bucket"
    type        = string
  }

  variable "enable_versioning" {
    description = "Enable versioning on the bucket"
    type        = bool
    default     = true
  }

  variable "sse_algorithm" {
    description = "Server-side encryption algorithm (AES256 or aws:kms)"
    type        = string
    default     = "AES256"
  }

  variable "force_destroy" {
    description = "Allow deletion of non-empty bucket"
    type        = bool
    default     = false
  }

  variable "tags" {
    description = "Tags to apply to the bucket"
    type        = map(string)
    default     = {}
  }

# modules/s3-bucket/main.tf
resource "aws_s3_bucket" "this" {
  bucket          = var.bucket_name
  force_destroy   = var.force_destroy
  tags            = var.tags
}

resource "aws_s3_bucket_versioning" "this" {
  bucket = aws_s3_bucket.this.id

  versioning_configuration {
    status = var.enable_versioning ? "Enabled" : "Suspended"
  }
}

resource "aws_s3_bucket_server_side_encryption_configuration" "this" {
  bucket = aws_s3_bucket.this.id

  rule {
    apply_server_side_encryption_by_default {
      sse_algorithm = var.sse_algorithm
    }
  }
}

resource "aws_s3_bucket_public_access_block" "this" {
```

```
bucket = aws_s3_bucket.this.id

block_public_acls      = true
block_public_policy    = true
ignore_public_acls     = true
restrict_public_buckets = true
}

# modules/s3-bucket/outputs.tf
output "bucket_id" {
  description = "Bucket name"
  value      = aws_s3_bucket.this.id
}

output "bucket_arn" {
  description = "Bucket ARN"
  value      = aws_s3_bucket.this.arn
}

output "bucket_domain_name" {
  description = "Bucket domain name"
  value      = aws_s3_bucket.this.bucket_domain_name
}

output "bucket_regional_domain_name" {
  description = "Bucket regional domain name"
  value      = aws_s3_bucket.this.bucket_regional_domain_name
}
```

3. Module Pattern: SQS Queue

```
# modules/sqs-queue/variables.tf
variable "queue_name" {
  description = "Name of the SQS queue"
  type       = string
}

variable "delay_seconds" {
  description = "Delay in seconds (0-900)"
  type       = number
  default    = 0
}

variable "message_retention_seconds" {
  description = "Message retention in seconds (60-1209600)"
  type       = number
  default    = 345600
}

variable "visibility_timeout_seconds" {
```

```
    description = "Visibility timeout in seconds (0-43200)"
    type        = number
    default     = 30
  }

  variable "receive_wait_time_seconds" {
    description = "Long poll wait time (0-20)"
    type        = number
    default     = 0
  }

  variable "tags" {
    description = "Tags to apply"
    type        = map(string)
    default     = {}
  }

  # modules/sqs-queue/main.tf
  resource "aws_sqs_queue" "this" {
    name                  = var.queue_name
    delay_seconds         = var.delay_seconds
    message_retention_seconds = var.message_retention_seconds
    visibility_timeout_seconds = var.visibility_timeout_seconds
    receive_wait_time_seconds = var.receive_wait_time_seconds
    tags                  = var.tags
  }

  # modules/sqs-queue/outputs.tf
  output "queue_id" {
    description = "Queue ID (URL)"
    value       = aws_sqs_queue.this.id
  }

  output "queue_arn" {
    description = "Queue ARN"
    value       = aws_sqs_queue.this.arn
  }

  output "queue_url" {
    description = "Queue URL"
    value       = aws_sqs_queue.this.url
  }
}
```

4. Module Pattern: DynamoDB Table

```
# modules/dynamodb-table/variables.tf
variable "table_name" {
  description = "Name of the DynamoDB table"
  type        = string
}
```

```
variable "hash_key" {
  description = "Partition key attribute name"
  type        = string
  default     = "pk"
}

variable "range_key" {
  description = "Sort key attribute name (optional)"
  type        = string
  default     = null
}

variable "billing_mode" {
  description = "Billing mode (PAY_PER_REQUEST or PROVISIONED)"
  type        = string
  default     = "PAY_PER_REQUEST"
}

variable "tags" {
  description = "Tags to apply"
  type        = map(string)
  default     = {}
}

# modules/dynamodb-table/main.tf
resource "aws_dynamodb_table" "this" {
  name           = var.table_name
  billing_mode   = var.billing_mode
  hash_key       = var.hash_key
  range_key      = var.range_key

  dynamic "attribute" {
    for_each = var.range_key != null ? [1, 2] : [1]
    content {
      name = attribute.key == 0 ? var.hash_key : var.range_key
      type = "S"
    }
  }

  attribute {
    name = var.hash_key
    type = "S"
  }

  tags = var.tags
}

# modules/dynamodb-table/outputs.tf
output "table_id" {
  value = aws_dynamodb_table.this.id
}

output "table_arn" {
```

```
    value = aws_dynamodb_table.this.arn
  }

  output "table_name" {
    value = aws_dynamodb_table.this.name
  }
```

5. Calling Reusable Modules

```
# main.tf - root configuration
module "orders_bucket" {
  source = "../modules/s3-bucket"

  bucket_name      = "app-orders-data"
  enable_versioning = true
  tags = {
    Environment = var.environment
    Service     = "orders"
  }
}

module "events_queue" {
  source = "../modules/sqs-queue"

  queue_name           = "app-order-events"
  delay_seconds        = 5
  message_retention_seconds = 604800

  tags = {
    Environment = var.environment
    Service     = "orders"
  }
}

module "orders_table" {
  source = "../modules/dynamodb-table"

  table_name = "app-orders"
  hash_key   = "orderId"

  tags = {
    Environment = var.environment
    Service     = "orders"
  }
}
```

6. Using Terraform Registry Modules

Public modules are available at registry.terraform.io:

```
module "vpc" {
  source = "terraform-aws-modules/vpc/aws"
  version = "5.0.0"

  name = "my-vpc"
  cidr = "10.0.0.0/16"

  azs          = ["us-east-1a", "us-east-1b"]
  private_subnets = ["10.0.1.0/24", "10.0.2.0/24"]
  public_subnets  = ["10.0.101.0/24", "10.0.102.0/24"]

  enable_nat_gateway = true
  enable_vpn_gateway = true

  tags = {
    Environment = "dev"
  }
}
```

7. Module Documentation

Every module should have a [README.md](#):

S3 Bucket Module

Creates an S3 bucket with versioning, encryption, and public access blocking.

Usage

```
```hcl
module "bucket" {
 source = "../modules/s3-bucket"

 bucket_name = "my-app-data"
 enable_versioning = true
 tags = {
 Environment = "dev"
 }
}
```
```

Inputs

| Name | Description | Type | Default | Required |
|-------------|-----------------------|--------|---------|----------|
| bucket_name | Name of the S3 bucket | string | - | yes |

```
enable_versioning	Enable versioning	`bool`	`true`	no
sse_algorithm	Encryption algorithm	`string`	`"AES256"`	no
force_destroy	Allow non-empty deletion	`bool`	`false`	no
tags	Resource tags	`map(string)`	`{}`	no
```

Outputs

```
Name	Description
bucket_id	Bucket name
bucket_arn	Bucket ARN
bucket_domain_name	Bucket domain name
```

8. Key Takeaways

- Reusable modules have **minimal required inputs** and **sensible defaults**
- Never hardcode project-specific values inside a module
- Expose all useful attributes as outputs
- Use **this** as the resource local name for consistency
- Document inputs and outputs in a README
- The Terraform Registry provides thousands of community modules
- Module version pinning (**version = "5.0.0"**) ensures reproducibility

example and exercise repo url

<https://github.com/eyu-se/terraform-training-material-with-projects>

Trainer – Eyuel M.